Nano2Predict · from the literature to a dataset of verified panels

1 · Literature

1,707 articles
Europe PMC · CC BY
nanocarriers
(no viral vectors)

Text and figure captions

Original figures

...

2 · Base v0.1 (frozen)

Descriptors
DLS · PDI · ζ

Spectra
FTIR · XRD · UV-vis · EDS…

Micrographs
TEM · SEM · AFM…

Every datum with its verbatim quote, verified by code (≥ 90 %)

3 · Figure quality control

1 Caption typology

panel labels (a, b, c…), plurals, overlaid curves

2 Visual classification

contact sheets; Claude and Codex blind to the caption; partial double reading

3 Image ↔ caption

agree → accepted
disagree → human queue

4 Five databases

single · pair · compound · overlaid · exclude

5 Panel separation

boxes on a grid;
geometric checks;
two-model agreement

6 Per-crop classifier

DINOv2 + human labels;
caption fragment per panel

7 Human verification

panel annotator (ground truth) · 200-panel sample with 95 % CI (Wilson)

4 · Output

Curated panels

micrograph ·
clean spectrum
id · class · technique
caption · box

Class and size models

Web Atlas and demo

Principle: precision over recall. Doubtful items are dropped; accepted items carry double evidence (caption + image).

Methodology inspired by EXSCLAIM!, MedICaT and FigEx2